

Technical Requirement Document

Here is the Technical Requirement Document (TRD) based on the provided Functional Requirement Document (FRD):

TR-DP-001: Load Customer and Order Data from CSV to Delta Tables

Related Functional Requirement(s): FR-DP-001

Objective: Implement a data ingestion job to load customer and order data from CSV files into Delta tables.

Target Cluster Configuration:

- Cluster Name: Data Ingestion Cluster
- Databricks Runtime Version: 10.4.x-scala2.12
- Node Type: Standard_DS3_v2
- Driver Node: 1 x Standard_DS3_v2
- Worker Nodes: 2-5 x Standard_DS3_v2 (autoscaling)
- Autoscaling: Enabled
- Auto Termination: 30 minutes
- Libraries Installed: delta-lake, spark-csv

Source Data Details:

- Dataset Name: Customer Data
- Location: /Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/customerdata
- Format: CSV
- Description: Customer data in CSV format
- Dataset Name: Order Data
- Location: /Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/orderdata
- Format: CSV
- Description: Order data in CSV format

Target Data Details:

- Output Name: Customer Delta Table
- Location: gen_ai_poc_databrickscoe.sdlc_wizard.customer_delta
- Format: Delta
- Description: Customer data in Delta format
- Output Name: Order Delta Table

- Location: gen_ai_poc_databrickscoe.sdlc_wizard.order_delta
- Format: Delta
- Description: Order data in Delta format

Job Flow / Pipeline Stages:

1. Read customer CSV data
2. Read order CSV data
3. Load customer data into Delta table
4. Load order data into Delta table

Data Transformations / Business Logic:

- Step: Read CSV data
- Description: Read customer and order CSV data
- Transformation Logic: Use spark-csv to read CSV data into DataFrames
- Step: Load data into Delta tables
- Description: Load customer and order data into Delta tables
- Transformation Logic: Use Delta Lake API to write DataFrames to Delta tables

Error Handling and Logging:

- Log errors and exceptions to a logging table
- Implement retry mechanism for failed data ingestion tasks

TR-DP-002: Clean Customer and Order Data

Related Functional Requirement(s): FR-DP-002

Objective: Implement data cleansing job to remove null and duplicate records from customer and order Delta tables.

Target Cluster Configuration:

- Cluster Name: Data Cleansing Cluster
- Databricks Runtime Version: 10.4.x-scala2.12
- Node Type: Standard_DS3_v2
- Driver Node: 1 x Standard_DS3_v2
- Worker Nodes: 2-5 x Standard_DS3_v2 (autoscaling)
- Autoscaling: Enabled
- Auto Termination: 30 minutes
- Libraries Installed: delta-lake

Source Data Details:

- Dataset Name: Customer Delta Table

- Location: gen_ai_poc_databrickscoe.sdlc_wizard.customer_delta
- Format: Delta
- Description: Customer data in Delta format
- Dataset Name: Order Delta Table
- Location: gen_ai_poc_databrickscoe.sdlc_wizard.order_delta
- Format: Delta
- Description: Order data in Delta format

Target Data Details:

- Output Name: Clean Customer Delta Table
- Location: gen_ai_poc_databrickscoe.sdlc_wizard.customer_clean
- Format: Delta
- Description: Cleaned customer data in Delta format
- Output Name: Clean Order Delta Table
- Location: gen_ai_poc_databrickscoe.sdlc_wizard.order_clean
- Format: Delta
- Description: Cleaned order data in Delta format

Job Flow / Pipeline Stages:

1. Remove null records from customer Delta table
2. Remove duplicate records from customer Delta table
3. Remove null records from order Delta table
4. Remove duplicate records from order Delta table

Data Transformations / Business Logic:

- Step: Remove null records
- Description: Remove rows with null values from customer and order Delta tables
- Transformation Logic: Use DataFrame API to filter out null records
- Step: Remove duplicate records
- Description: Remove duplicate records from customer and order Delta tables
- Transformation Logic: Use DataFrame API to remove duplicate records

Error Handling and Logging:

- Log errors and exceptions to a logging table
- Implement retry mechanism for failed data cleansing tasks

TR-DP-003: Create ordersummary Table and Load Joined Customer and Order Data

Related Functional Requirement(s): FR-DP-003

Objective: Implement a data processing job to create an "ordersummary" table and load joined customer and order data into it using SCD type 2.

Target Cluster Configuration:

- Cluster Name: Data Processing Cluster
- Databricks Runtime Version: 10.4.x-scala2.12
- Node Type: Standard_DS3_v2
- Driver Node: 1 x Standard_DS3_v2
- Worker Nodes: 2-5 x Standard_DS3_v2 (autoscaling)
- Autoscaling: Enabled
- Auto Termination: 30 minutes
- Libraries Installed: delta-lake

Source Data Details:

- Dataset Name: Clean Customer Delta Table
- Location: gen_ai_poc_databrickscoe.sdlc_wizard.customer_clean
- Format: Delta
- Description: Cleaned customer data in Delta format
- Dataset Name: Clean Order Delta Table
- Location: gen_ai_poc_databrickscoe.sdlc_wizard.order_clean
- Format: Delta
- Description: Cleaned order data in Delta format

Target Data Details:

- Output Name: ordersummary Table
- Location: gen_ai_poc_databrickscoe.sdlc_wizard.ordersummary
- Format: Delta
- Description: Joined customer and order data in Delta format using SCD type 2

Job Flow / Pipeline Stages:

1. Check if "ordersummary" table exists
2. Create "ordersummary" table if it does not exist
3. Join customer and order data
4. Load joined data into "ordersummary" table using SCD type 2

Data Transformations / Business Logic:

- Step: Join customer and order data
- Description: Join customer and order data on CustId
- Transformation Logic: Use DataFrame API to join customer and order data

- Step: Load data into "ordersummary" table
- Description: Load joined data into "ordersummary" table using SCD type 2
- Transformation Logic: Use Delta Lake API to implement SCD type 2 logic

Error Handling and Logging:

- Log errors and exceptions to a logging table
- Implement retry mechanism for failed data processing tasks

TR-DP-004: Update ordersummary SCD Type 2 Table on Customer Data Change

Related Functional Requirement(s): FR-DP-004

Objective: Implement a data processing job to update the "ordersummary" SCD type 2 table whenever there is a change in the customer table.

Target Cluster Configuration:

- Cluster Name: Data Processing Cluster
- Databricks Runtime Version: 10.4.x-scala2.12
- Node Type: Standard_DS3_v2
- Driver Node: 1 x Standard_DS3_v2
- Worker Nodes: 2-5 x Standard_DS3_v2 (autoscaling)
- Autoscaling: Enabled
- Auto Termination: 30 minutes
- Libraries Installed: delta-lake

Source Data Details:

- Dataset Name: Customer Delta Table
- Location: gen_ai_poc_databrickscoe.sdmc_wizard.customer_delta
- Format: Delta
- Description: Customer data in Delta format

Target Data Details:

- Output Name: ordersummary Table
- Location: gen_ai_poc_databrickscoe.sdmc_wizard.ordersummary
- Format: Delta
- Description: Updated "ordersummary" table using SCD type 2

Job Flow / Pipeline Stages:

1. Detect changes in customer table
2. Update corresponding records in "ordersummary" table

Data Transformations / Business Logic:

- Step: Detect changes in customer table
- Description: Detect changes in customer data
- Transformation Logic: Use DataFrame API to compare current and previous customer data
- Step: Update "ordersummary" table
- Description: Update corresponding records in "ordersummary" table using SCD type 2 logic
- Transformation Logic: Use Delta Lake API to implement SCD type 2 logic

Error Handling and Logging:

- Log errors and exceptions to a logging table
- Implement retry mechanism for failed data processing tasks

TR-DP-005: Create customeraggregatespend Table and Load Aggregated Spend Data**Related Functional Requirement(s): FR-DP-005**

Objective: Implement a data processing job to create a "customeraggregatespend" table and load aggregated spend data from the "ordersummary" table.

Target Cluster Configuration:

- Cluster Name: Data Processing Cluster
- Databricks Runtime Version: 10.4.x-scala2.12
- Node Type: Standard_DS3_v2
- Driver Node: 1 x Standard_DS3_v2
- Worker Nodes: 2-5 x Standard_DS3_v2 (autoscaling)
- Autoscaling: Enabled
- Auto Termination: 30 minutes
- Libraries Installed: delta-lake

Source Data Details:

- Dataset Name: ordersummary Table
- Location: gen_ai_poc_databrickscoe.sdlc_wizard.ordersummary
- Format: Delta
- Description: Joined customer and order data in Delta format using SCD type 2

Target Data Details:

- Output Name: customeraggregatespend Table
- Location: gen_ai_poc_databrickscoe.sdlc_wizard.customeraggregatespend
- Format: Delta

- Description: Aggregated spend data in Delta format

Job Flow / Pipeline Stages:

1. Check if "customeraggregatespend" table exists
2. Create "customeraggregatespend" table if it does not exist
3. Aggregate spend data from "ordersummary" table
4. Load aggregated data into "customeraggregatespend" table

Data Transformations / Business Logic:

- Step: Aggregate spend data
- Description: Aggregate spend data from "ordersummary" table
- Transformation Logic: Use DataFrame API to aggregate spend data
- Step: Load data into "customeraggregatespend" table
- Description: Load aggregated data into "customeraggregatespend" table
- Transformation Logic: Use Delta Lake API to write aggregated data to Delta table

Error Handling and Logging:

- Log errors and exceptions to a logging table
- Implement retry mechanism for failed data processing tasks